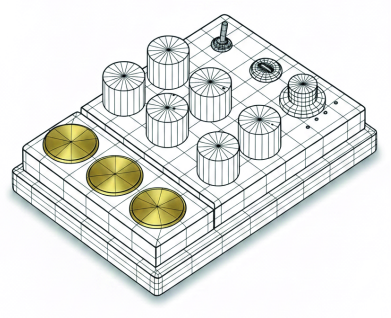
// generative instrument

Modal_OS

diceFriend / quick start



Ringing strings, metal bars, hollow caves and clustered resonators.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: self-exciting stereo modal bank. **TS:** three struck resonators.

TS: T1-T3 strike separate modal voices; hold for longer resonant gestures.

// KNOBS / AS

P	NORMAL	SHIFT
1	Tempo / MIDI transpose*	Drive / excitation
2	Left root pitch	LFSR rate
3	Right root pitch	Resonator density
4	Feedback	Detune / stereo
5	Exciter chaos	LFSR density
6	Decay	Modal mix

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 pitch	Exciter chaos
2	T2 pitch	Exciter level
3	T3 pitch	Feedback
4	T1 decay	Drive
5	T2 decay	Detune
6	T3 decay	Fold / strike character

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE in AS: randomize. **Tap in TS:** Strings / Bars / Cave / Poly. **SHIFT + tap:** randomize AS or TS timbre. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: trigger / clock. **CV2:** modulation; repeated pulses can become clock / triggers. CV1 can strike all TS voices.
Banana: 0-9 V; common GND.

// USB MIDI

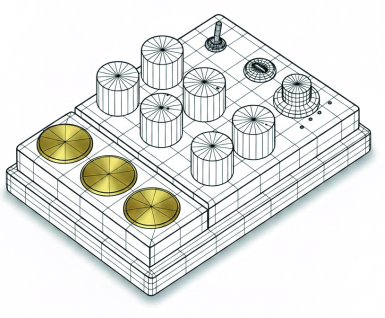
IN / OUT: notes and clock. **AS CC:** 20 = left pitch, 21 = speed, 22 = right pitch, 23-25 = feedback / chaos / decay; 26-31 = SHIFT. **TS:** CC52-57 normal, CC58-63 SHIFT.

*With external clock, normal P1 transposes the second MIDI-output layer. Raise decay and feedback gradually for sustained resonance.

// generative instrument

Run_OS

diceFriend / quick start



Raw rungler sequences, folded oscillators and four nonlinear touch engines.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: BiXo runs itself. **TS:** Trigfeto / Feton / Artefakto / Bulgo.

AS pads: T1 boosts A; T2 boosts B; T3 adds cross-bleed. All three freeze stepping. **TS:** three playable voices.

// KNOBS / AS

P	NORMAL	SHIFT
1	Tempo / MIDI transpose*	Rungler bias
2	Oscillator A pitch	Oscillator B pitch
3	Oscillator A decay	Oscillator B decay
4	Rungler chaos	Cross-modulation ratio
5	Cross-bleed	Feedback low-pass
6	Oscillator A fold	Oscillator B fold

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 pitch	T1 tone / filter
2	T2 pitch	T2 tone / filter
3	T3 pitch	T3 tone / filter
4	T1 decay	T1 fold / feedback
5	T2 decay	T2 fold / feedback
6	T3 decay	T3 fold / feedback

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE in AS: randomize the selected layer. **In TS:** next engine. **SHIFT + DICE 3 s:** randomize SHIFT parameters. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: trigger / clock. **CV2:** modulation; repeated pulses can become clock / triggers. In TS, CV1 triggers the three voices together.
Banana: 0-9 V; common GND.

// USB MIDI

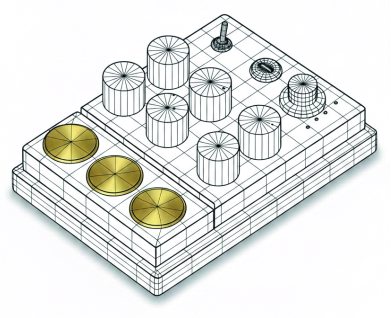
IN: notes, clock; CC74 modulates TS. **OUT:** generated / pad notes, pressure, CV2 pitch bend and local clock. **Panel:** CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

*With external clock, normal P1 transposes the second MIDI-output layer. Artefakto uses SHIFT P4-P6 for chaos / feedback depth.

// generative instrument

NonLin_OS

diceFriend / quick start



Lorenz and Dadras attractors become melodies, resonant pings and nonlinear textures.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: scale-quantized attractor motion. **TS:** Duffing / Granular / LFSR / Rossler.

AS: pads reshape the attractor; all three freeze stepping. **TS:** three voices with individual pitch and decay.

// KNOBS / AS

P	NORMAL	SHIFT
1	Tempo / MIDI transpose*	Scale selection
2	Root pitch	Pitch range
3	Lorenz width / Dadras B	Fold
4	Chaos / Dadras C	Bleed / texture
5	Filter / Dadras D	Feedback low-pass
6	Decay	Ping / fold intensity

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 pitch	T1 tone / filter
2	T2 pitch	T2 tone / filter
3	T3 pitch	T3 tone / filter
4	T1 decay	T1 fold / feedback
5	T2 decay	T2 fold / feedback
6	T3 decay	T3 fold / feedback

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE in AS: new scene and scale. **SHIFT + tap in AS:** Lorenz / Dadras. **Tap in TS:** next engine. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: trigger / clock. **CV2:** modulation; repeated pulses can become clock / triggers. CV1 triggers the TS group.
Banana: 0-9 V; common GND.

// USB MIDI

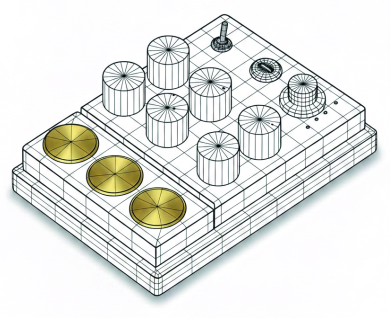
IN: notes, clock; CC74 modulates TS. **OUT:** generated / pad notes, pressure, CV2 pitch bend; attractor movement on CC16-18.

*With external clock, normal P1 transposes the second MIDI-output layer. Start with low chaos and fold, then raise ping intensity.

// generative instrument

Mult_OS

diceFriend / quick start



A walking three-voice cluster with four engines and four mutation depths.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: autonomous pitch walk. **TS:** Wave Cluster / Glass FM / Chip Swarm / Bounce.

TS: T1-T3 play their own pitch and decay. Holds let the voices develop.

// KNOBS / AS

P	NORMAL	SHIFT
1	Tempo / MIDI transpose*	Scale / voice 1 detune
2	Base pitch	Walk range / voice 2 detune
3	Harmonic spread / PM	Fold / voice 3 detune
4	Feedback / cross-mod	Modulation depth
5	Motion	Feedback damping
6	Release	Space / stereo

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 pitch	Harmonic spread
2	T2 pitch	Phase fold
3	T3 pitch	Tone / brightness
4	T1 decay	PM depth
5	T2 decay	Feedback
6	T3 decay	Motion / stereo

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: mutate AS or select the next TS engine. **SHIFT + tap:** next depth + mutation: Safe / Musical / Chaos / Wild. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: trigger / clock. **CV2:** modulation; repeated pulses can become clock / triggers.
Banana: 0-9 V; common GND.

// USB MIDI

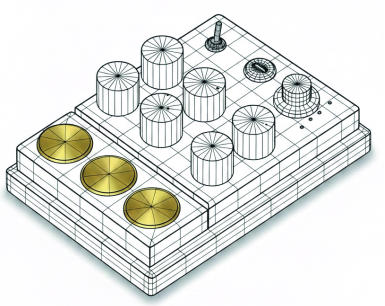
IN: notes and MIDI Clock; CC74 = CV2 modulation. **OUT:** local notes, panel CC and internal / CV clock. **Panel:** CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

*With external clock, normal P1 transposes the second MIDI-output layer. Start resets the walk; Stop clears incoming notes.

// generative instrument

GenTex_OS

diceFriend / quick start



A self-referencing stereo texture loop: feedback, folding, filtering and micro-delay.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: evolving feedback texture. **TS:** one shared gated texture; pad pitches blend.

AS: T1 Freeze / T2 Smear / T3 Spike. **TS:** T1 adds fold, T2 delay movement, T3 filter movement. Holds deepen the gesture.

// KNOBS / AS

P	NORMAL	SHIFT
1	Feedback	Pre-gain
2	Fold	Fold asymmetry
3	Delay time	Delay smear
4	Chaos injection	Bit depth
5	Filter cutoff	Filter mode
6	Filter resonance	Output saturation

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 pitch contribution	Pre-gain
2	T2 pitch contribution	Fold asymmetry
3	T3 pitch contribution	Delay smear
4	Feedback	Bit depth
5	Texture / brightness	Filter mode
6	Delay / resonance	Output saturation

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: randomize both layers. **SHIFT + DICE 3 s:** randomize without changing mode. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: continuous fold / chaos modulation; pulses can trigger. **CV2:** pulse / clock detection and texture macro gestures.
Banana: 0-9 V; common GND.

// USB MIDI

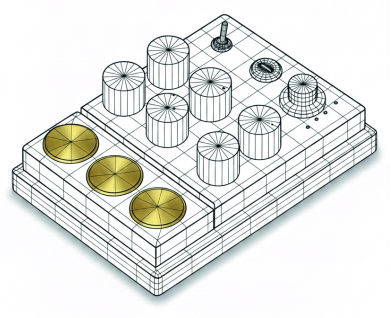
IN: notes open the TS gate and change timbre; Note Off closes it. MIDI Clock and musical MIDI Thru are supported. **OUT:** texture notes, energy aftertouch and CV pitch bend.

Small feedback changes can transform the whole texture. TS uses a shared engine; it is not three independent resonators.

// generative instrument

Kac_OS

diceFriend / quick start



Delta-sigma / Lorenz phrases and an eight-region FM/PM morph matrix.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: evolving delta-sigma phrases. **TS:** KACC / Metal / Fold / Grind.

AS: T1 freezes pitch; T2 widens the walk; T3 adds glitch / fold. **TS:** three voices; pads have priority over incoming MIDI.

// KNOBS / AS

P	NORMAL	SHIFT
1	Tempo / MIDI transpose*	Fine ratio / detune
2	Modulator ratio	Cross-feedback
3	FM / PM depth	Mutation probability
4	Wavefolder	Pattern length: 2-16
5	Harmonic stretch	Smoothing / pitch range
6	Morph position	Saturation

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 pitch	FM / PM depth
2	T2 pitch	Phase grit / feedback
3	T3 pitch	Modulator ratio
4	T1 decay	Wavefolder
5	T2 decay	Phase flip
6	T3 decay	KACC morph

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: randomize AS while preserving tempo, or select the next TS engine. **SHIFT + DICE 3 s:** randomize controls. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: trigger / clock. **CV2:** modulation; repeated pulses can become clock / triggers. In TS, external triggers allocate voices.
Banana: 0-9 V; common GND.

// USB MIDI

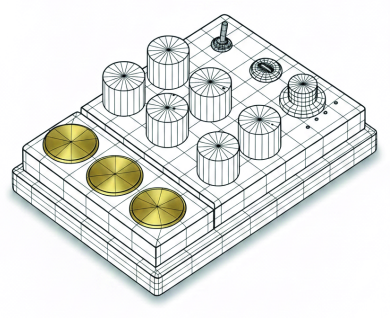
IN: notes with three-voice TS allocation and Note Off.
OUT: AS notes; TS pads on channels 1 / 2 / 3; pressure and CV2 pitch bend. Musical MIDI Thru.
Panel: CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

*With external clock, normal P1 transposes the second MIDI-output layer. TS pitch knobs detune MIDI by about +/-1 octave.

// generative instrument

Pulsar_OS

diceFriend / quick start



Dual pulsar orbits: isolated impulses, folded rhythms and feedback clouds.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: dual pulsar orbit. **TS:** Classic / Cross-Feedback / Folded / Feedback Cloud.

AS: T1 freezes orbit / steps; T2 adds bursts and smear; T3 adds nonlinear pulse contrast. **TS:** three gated voices.

// KNOBS / AS

P	NORMAL	SHIFT
1	Tempo / density / transpose*	Stereo / audio-rate range
2	Pulse width	Cross-feedback
3	Feedback	Pattern mutation
4	Nonlinear shape	Pattern length: 2-16
5	Drift / chaos	Smoothing / orbit
6	Tone / filter	Output drive

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 pitch	Pulse width
2	T2 pitch	Feedback
3	T3 pitch	Nonlinear shape
4	T1 decay	Stereo spread
5	T2 decay	Body / resonance
6	T3 decay	Output drive

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: randomize AS without changing tempo; in TS, randomize the scene and choose an engine. **SHIFT + DICE 3 s:** randomize / reset. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: trigger / clock. **CV2:** modulation; repeated pulses can become clock / triggers.
Banana: 0-9 V; common GND.

// USB MIDI

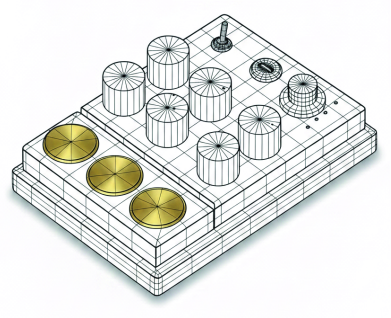
IN: notes with three-voice TS allocation; Clock. **OUT:** TS pads on channels 1 / 2 / 3, AS notes, CC16-18, CV2 pitch bend. Musical MIDI Thru. **Panel:** CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

*With external clock, normal P1 transposes the second MIDI-output layer. Low density and narrow width give separate impulses; high settings approach drones.

// generative instrument

Blip_OS

diceFriend / quick start



Coupled oscillators, shift registers and resonant peaks move between clicks and ringing drones.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: continuously active core. **TS:** the same shared core opens from touch or triggers.

T1: rate movement. **T2:** rungler instability. **T3:** resonant-peak motion. In TS, any pad opens the sound; holds deepen it.

// SIX KNOBS / TWO LAYERS

P	NORMAL	SHIFT
1	Clock / rungler A*	Oscillator A rate
2	Oscillator B relation	Rungler B
3	Resonant peak 1	Sample & hold A
4	Resonant peak 2	Sample & hold B
5	Spread / excitation	Register source balance
6	Gain / grit	Resonance / stereo

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: randomize. **SHIFT + tap:** deeper variation. **DICE 1.8 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: trigger / clock. **CV2:** modulation; repeated pulses can become clock / triggers. CV1 / CV2 pulses produce MIDI notes 36 / 37.
Banana: 0-9 V; common GND.

// USB MIDI

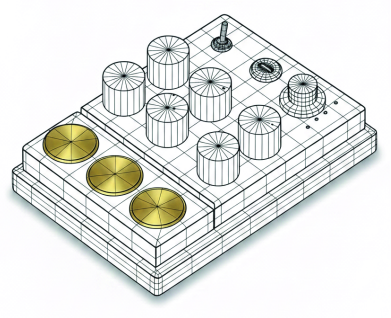
IN: notes retune and strike; Clock drives strikes. **OUT:** CV notes, CV2 CC74 and internal Clock. Incoming musical / realtime MIDI is passed through. **Panel:** CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

*Under external clock, normal P1 controls the second MIDI-output transpose. Raise normal P5 and SHIFT P6 for ringing textures.

// generative instrument

Kac_OS Bass

diceFriend / quick start



Delta-sigma / Lorenz phrases and an eight-region FM/PM morph matrix. Audio is two octaves lower.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: evolving delta-sigma phrases. **TS:** KACC / Metal / Fold / Grind.

AS: T1 freezes pitch; T2 widens the walk; T3 adds glitch / fold. **TS:** three voices; pads have priority over incoming MIDI.

// KNOBS / AS

P	NORMAL	SHIFT
1	Tempo / MIDI transpose*	Fine ratio / detune
2	Modulator ratio	Cross-feedback
3	FM / PM depth	Mutation probability
4	Wavefolder	Pattern length: 2-16
5	Harmonic stretch	Smoothing / pitch range
6	Morph position	Saturation

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 pitch	FM / PM depth
2	T2 pitch	Phase grit / feedback
3	T3 pitch	Modulator ratio
4	T1 decay	Wavefolder
5	T2 decay	Phase flip
6	T3 decay	KACC morph

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: randomize AS while preserving tempo, or select the next TS engine. **SHIFT + DICE 3 s:** randomize controls. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: trigger / clock. **CV2:** modulation; repeated pulses can become clock / triggers. In TS, external triggers allocate voices.
Banana: 0-9 V; common GND.

// USB MIDI

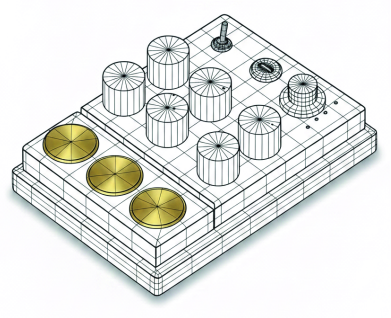
IN: notes with three-voice TS allocation and Note Off.
OUT: AS notes; TS pads on channels 1 / 2 / 3; pressure and CV2 pitch bend. Musical MIDI Thru.
Panel: CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

*With external clock, normal P1 transposes the second MIDI-output layer. TS pitch knobs detune MIDI by about +/-1 octave. Bass audio is shifted down two octaves, with a 20 Hz floor.

// generative instrument

PERC_OS

diceFriend / quick start



A touch percussion looper: kick, snare, hi-hat, rhythm mutation and four synthesis engines.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

Loop + Touch: one operating model. A 32-step loop spans two bars of 4/4.

T1: kick. **T2:** snare. **T3:** hi-hat. Tap to play and record. Hold a pad 3 s to clear its track.

// SIX KNOBS / TWO LAYERS

P	NORMAL	SHIFT
1	Kick character	Tempo: 20-240 BPM
2	Snare character	Chaos / mutation
3	Hi-hat character	Feel: quantize > swing
4	Kick decay	Compression
5	Snare decay	Fracture
6	Hi-hat decay	Resonant space

SHIFT: up = second layer; down = normal. **Pickup:** After switching layers, move the knob to take control. Touching a pad also unlocks its sound-control pair.

// DICE / SHIFT

Tap DICE: new sound, same beat. **SHIFT + tap:** rhythm mutation. **DICE 3 s:** next engine. **DICE + pad:** randomize / audition that voice. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1 / CV2: clock inputs only. One pulse advances one sixteenth; an empty track stays silent.
Banana: 0-9 V; common GND.

// USB MIDI

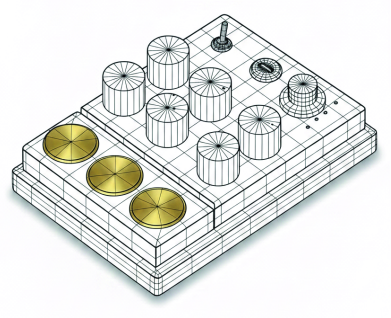
IN: Clock / transport; notes are ignored. CC20-25 = SHIFT globals; CC26-31 = normal sound controls. **OUT:** Clock and panel CC; loop hits do not send notes.

Start with SHIFT P1 tempo, then record the three pads. Hold DICE beyond 0.7 s to raise chaos; SHIFT + DICE 3 s then release commits rhythm mutation.

// generative instrument

DriftScape_OS

diceFriend / quick start



A wandering low body, two wind-chime melodies and long tonal reflections.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: autonomous ambient landscape. **TS:** three dedicated resonators with individual decay.

AS: T1 Freeze / T2 Dissolve / T3 Bloom. **TS:** T1-T3 strike their own resonators; hold for a longer gesture.

// KNOBS / AS

P	NORMAL	SHIFT
1	Melody A / B balance	MIDI transpose / root
2	Mutation	Drift
3	Body decay	Stereo spread
4	Reverb / space	Tempo: 20-180 BPM
5	Chime decay	Chime register
6	Chime timbre	Body / chime balance

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 pitch	Resonance / Q
2	T2 pitch	Stereo spread
3	T3 pitch	Brightness
4	T1 decay	Resonator character
5	T2 decay	Reserved
6	T3 decay	Gentle fold

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: new landscape. **SHIFT + DICE 3 s:** deeper randomization. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: trigger / clock. **CV2:** modulation; repeated pulses can become clock / triggers. CV2 also modulates space and chime timbre; steady level sends CC74.
Banana: 0-9 V; common GND.

// USB MIDI

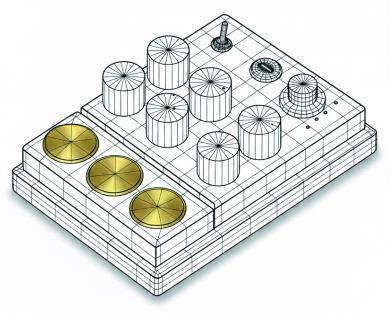
IN: notes set the tonal center and excite AS / TS; Clock. **OUT:** pad / CV notes and internal / CV Clock. SHIFT P1 transposes generated MIDI, not incoming notes.

For a calm field, lower Mutation and raise Body Decay. Center Melody Balance to hear both chime layers. TS SHIFT P5 has no assigned function.

// generative instrument

CIG_OS

diceFriend / quick start



An impulse core with a retained-memory layer: folding, scanning and uneven clock motion.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: self-moving impulse / memory core. **TS:** the shared core is gated by touch and triggers.

T1: rate / folding. **T2:** rungler / feedback. **T3:** resonant peaks / tone. Holds increase their influence.

// SIX KNOBS / TWO LAYERS

P	NORMAL	SHIFT
1	Clock / post fold*	Oscillator A rate
2	Oscillator B relation	Post feedback
3	Resonant peak 1	Post tone
4	Resonant peak 2	Retention amount
5	Spread / excitation	Retention scan
6	Gain / grit / resonance	Jitter / clock unevenness

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: randomize. **SHIFT + tap:** deeper variation. **DICE 1.8 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: trigger / clock. **CV2:** modulation; repeated pulses can become clock / triggers. CV edges generate notes 36 / 37; steady CV2 sends CC74, bend and pressure.
Banana: 0-9 V; common GND.

// USB MIDI

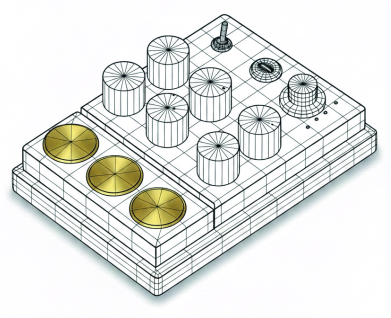
IN: notes and Clock. **OUT:** CV notes, panel CC, internal / CV Clock. Incoming USB messages are not echoed. **Panel:** CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

*With external clock, normal P1 transposes the second MIDI-output layer. SHIFT P6 bends internal timing; Retention + Scan reveal past material.

// generative instrument

4S_OS

diceFriend / quick start



Four interacting slopes collide, exchange energy and form breathing cybernetic drones.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: continuously active four-slope network. **TS:** all pads open and reshape the same shared stereo network.

T1: rate / folding. **T2:** coupling / feedback. **T3:** spectrum / instability. Holds deepen each disturbance.

// SIX KNOBS / TWO LAYERS

P	NORMAL	SHIFT
1	Boundaries / clock*	Slope A rate
2	Slope B rate	Drift
3	Slope C rate	Shape / fold
4	Slope D rate	Spectrum / smoothing
5	Coupling	Feedback
6	Rise / fall asymmetry	Stereo spread

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: randomize. **SHIFT + tap:** deeper mutation. **DICE 1.8 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: trigger / clock. **CV2:** modulation; repeated pulses can become clock / triggers. Each external step changes the temporary network configuration. **Banana:** 0-9 V; common GND.

// USB MIDI

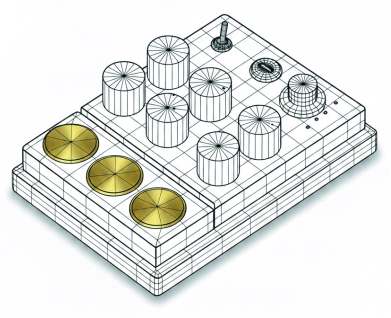
IN: notes create mutations; note number seeds variation instead of conventional transposition. Clock steps mutate. **OUT:** CV notes 36 / 37, CV2 CC74 and local / CV Clock. **Panel:** CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

*With external clock, normal P1 controls the second MIDI-output transpose. Start with Coupling below halfway; raise it gradually to hear moving boundaries.

// generative instrument

Chimera_OS

diceFriend / quick start



Runglers, FM/PM, pulsars and low-pass gates share a cybernetic sound organism.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: one organism, with one event per 2-5 source steps. **TS:** three independent voices without that divider.

AS: T1 Freeze / T2 Invert / T3 Rupture; new touches strike. **TS:** each pad gates its own voice.

// KNOBS / AS

P	NORMAL	SHIFT
1	Root pitch	Coupling / feedback
2	Modulator ratio	Memory time
3	Ecology: osc > pulsar	Mutation
4	Fold	LPG decay
5	Aperture / LPG brightness	Homeostasis
6	Density / base clock	Space

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 pitch	Coupling
2	T2 pitch	Memory
3	T3 pitch	Mutation
4	T1 decay	Fold
5	T2 decay	Aperture
6	T3 decay	Space

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: randomize all 12 active-mode parameters and strike. **SHIFT + tap:** reset / resume without panel randomization. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: edge triggers. **CV2:** pitch modulation and edges. AS divides CV events 2-5; TS triggers successive voices directly.
Banana: 0-9 V; common GND.

// USB MIDI

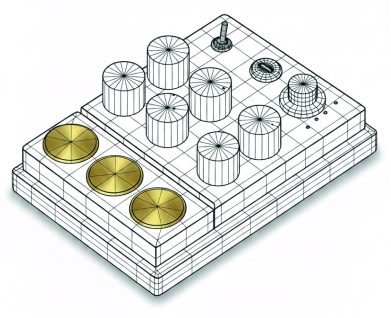
IN: channel 1 notes and Clock; AS divides events 2-5. CC120 stops; CC123 releases / resets. **OUT:** local notes, CV2 CC74 and panel CC. **Panel:** CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

Start with low Density and modest Coupling. Incoming MIDI Clock disables local Clock Out until a local short DICE; release pads after Stop before playing again.

// generative instrument

Ping_OS

diceFriend / quick start



Coupled resonators turn clocked impacts into pings, melodic fragments and feedback gestures.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: coupled autonomous pings. **TS:** three resonators, each with its own attack and decay.

TS: T1 plays A, T2 plays C, T3 plays B. Pitch, decay, colour and attack are individually mapped.

// KNOBS / AS

P	NORMAL	SHIFT
1	Tempo / master pitch*	A > B coupling
2	A pitch / interval*	B > A coupling
3	B pitch / interval*	Cross-FM / steps
4	Chaos / step depth	Threshold
5	Coupling / decay	Feedback
6	Event density	Drive

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 / A pitch	A colour
2	T2 / C pitch	C colour
3	T3 / B pitch	B colour
4	A decay	A attack
5	C decay	C attack
6	B decay	B attack

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: randomize the normal layer and audition. **SHIFT + tap:** randomize SHIFT. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: trigger / clock. **CV2:** modulation; repeated pulses can become clock / triggers. CV takes timing priority.
Banana: 0-9 V; common GND.

// USB MIDI

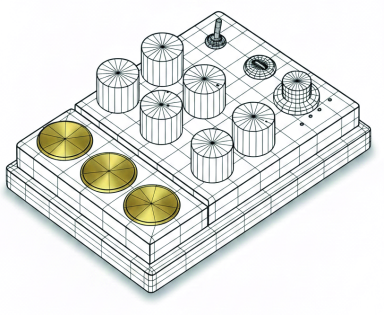
IN: notes trigger AS or the TS group (root / +7 / +12); CC74 = CV2. No CC20-31 panel receiver.
OUT: local ping notes and internal / CV Clock.

*With external clock, normal P1 becomes master pitch and P2/P3 set intervals from it. Note Off releases pitch override; MIDI Stop does not silence the instrument.

// generative instrument

Flux_OS

diceFriend / quick start



Three FM voices, remembered timbres and rhythms that stretch between pulses and evolving phrases.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: irregular autonomous FM phrases. **TS:** three voices, with Strike / Hold / Bloom articulations.

AS: T1 accents depth; T2 shifts ratios; T3 freezes memory / motion. **TS:** each pad has its own pitch and decay.

// KNOBS / AS

P	NORMAL	SHIFT
1	Fundamental pitch	FM topology
2	FM ratio	Detune
3	FM depth	Feedback
4	Decay	Contour
5	Rate: 30-300 BPM*	Body resonator
6	Motion / irregularity	Stereo space

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 pitch	FM ratio
2	T2 pitch	FM depth
3	T3 pitch	Topology
4	T1 decay	Feedback
5	T2 decay	Contour
6	T3 decay	Detune

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE in AS: randomize normal layer, keep Rate. **SHIFT + tap:** randomize SHIFT. **Tap in TS:** Strike / Hold / Bloom. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: AS step or three TS triggers. **CV2:** Depth modulation; repeated edges can become clock. **Banana:** 0-9 V; common GND.

// USB MIDI

IN: notes, Clock; bend +/-2 semitones; pressure / CC1 / CC74 raise Depth. **OUT:** local notes, CC and internal / CV Clock. **Panel:** CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

*With external clock, AS normal P5 selects /4, /2, x1, x2 or x4. TS pad pressure does not change timbre; Body and Space belong to AS.

// generative instrument

Donut_OS

diceFriend / quick start

Two voices walk a wrapping pitch grid through six tunings, folding, granular memory and feedback.

START: USB-C power + 3.5 mm stereo audio out. Begin with the listening level low.

T1 taps tempo. T2 pressure adds brightness and grain intensity. T3 pressure lengthens decay.

// SIX KNOBS / TWO LAYERS

P	NORMAL	SHIFT
1	Scale: 6 tuning maps	Density: audible-step probability
2	Interval span	Chaos: grain / register instability
3	Width: 2-32 cells	Mutation: direction / spectrum
4	Height: 2-32 cells	Feedback: motion / fold / colour
5	Bias: X / Y motion	Inertia: retain axis / direction
6	Root: C1-C4	Route: body / texture / decay

// TUNINGS / ROUTE

Tunings: 12 EDO / 19 EDO / 31 EDO / Hexany A / Hexany B / 5-limit JI. ROUTE selects how grid position, internal chaos and CV2 reach body / folding, texture / FM or decay.
SHIFT: up = second layer; down = normal. Pickup: after changing layers, using DICE or receiving CC, move a knob to take control.

// DICE / LOOP

Tap DICE: new world and immediate strike. SHIFT + tap DICE: related mutation preserving scale, root, grid and route. Hold DICE 450 ms: loop up to 8 recent events; release to continue from the recalled position. SHIFT + DICE 10 s: bootloader.

// CLOCK / TRANSPORT

CV > MIDI > internal. MIDI may take over 20 s after CV; internal returns after 30 s without external activity. MIDI Clock is 24 PPQN. Start / Continue run; Stop silences MIDI-clocked motion. Note On can still strike after Stop.

// CV IN

CV1: priority step / clock. CV2: continuous ROUTE modulation or secondary trigger / clock; CV1 blocks extra CV2 steps. Banana: 0-9 V; common GND.

// USB MIDI

IN ch 1: Note On forces a step; note number does not retune. CC20-31 = panel, CC74 / Aftertouch = expression. OUT: voices A / B on ch 1 / 2 with Pitch Bend, Note On / Off and internal / CV Clock.

// FIRST PATCH

Choose 19 EDO, medium WIDTH / HEIGHT, high DENSITY and low CHAOS / MUTATION / FEEDBACK. Tap T1 twice. Raise CHAOS, then hold DICE when an eight-step phrase appears.

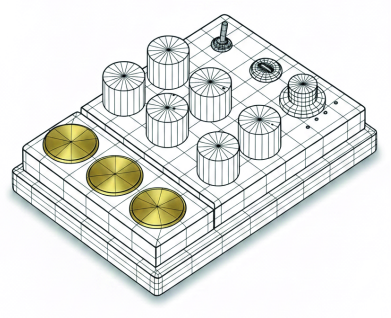
// LEDS

Layer LEDs show NORMAL / SHIFT. Lower LEDs: step activity; CV2 or loop blink; left audio; right audio.

// generative instrument

RELAB_OS

diceFriend / quick start



Five coupled mechanisms bend time, share energy and rework the remains of earlier events.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: autonomous time organism. **TS:** pads open Torque / Sheet / Friction; MIDI reaches all five mechanisms.

AS: T1 Gather / T2 Shear / T3 Avalanche. **TS:** hold T1-T3 to keep their respective regions open.

// SIX KNOBS / TWO LAYERS

P	NORMAL	SHIFT
1	Density / micro-cut rate	Root pitch
2	Relation	Cross-modulation
3	Erase	Decay / overlap
4	Material balance	Slip range
5	Memory	Memory scan
6	Pressure / avalanche	Drive / repeats / crushing

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: strong mutation with Density preserved. **SHIFT + tap:** wider catastrophic mutation. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: priority trigger / clock. **CV2:** Erase / Pressure modulation; fast edges trigger Arc / Friction / Wreckage without taking transport.
Banana: 0-9 V; common GND.

// USB MIDI

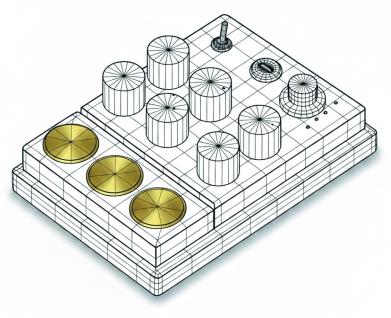
IN: channel 1 rotates voices; channels 2-6 address mechanisms 1-5. Notes, transport, Clock, CC74. **OUT:** melodic lines on channels 1 / 2; local / CV Clock.
Panel: CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

Start with Density and Memory near halfway, Erase low. Reduce Pressure, Drive and Cross-mod first after an extreme mutation.

// generative instrument

Pull_OS

diceFriend / quick start



Granulated pulse trains, formants, rungler motion and a torn stereo memory.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: autonomous granular pulse organism. **TS:** three independent voices with separate pitch and decay.

AS: T1 Freeze / T2 Rupture / T3 Phase Space. **TS:** T1-T3 gate voices; SHIFT + held pad adds rupture.

// KNOBS / AS

P	NORMAL	SHIFT
1	Grain emission rate	Grain overlap
2	Grain duration	Memory
3	Formant frequency	Phaser
4	Pulse shape	Wavefold
5	Chaos	Mutation
6	Tempo / release	Stereo space

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 pitch	Duration / overlap
2	T2 pitch	Formant
3	T3 pitch	Shape / fold
4	T1 decay	Memory / phaser
5	T2 decay	Chaos / mutation
6	T3 decay	Space / clock rate

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: mutate the current layer. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: priority step / trigger. **CV2:** smoothed continuous modulation across pulse, formant and effects; sends CC74.
Banana: 0-9 V; common GND.

// USB MIDI

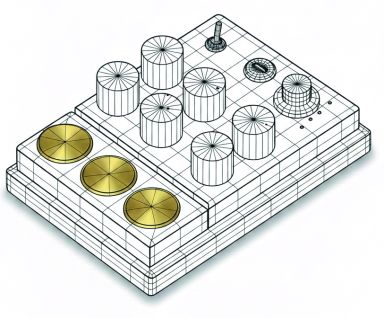
IN: channel 1 notes, transport, 24 PPQN Clock.
OUT: local / CV / pad notes, panel CC and internal / CV Clock. **Panel:** CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

CV has clock priority; MIDI may take over after 20 s without CV, internal after 30 s without external timing. Increase Memory and Fold gradually.

// generative instrument

Aether_OS

diceFriend / quick start



A melodic attractor and granular pulsar weave masked impulses through fractured stereo memory.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: scale-based attractor melody. **TS:** three gated voices with separate pitch and decay.

AS: T1 freezes rungler / memory writes; T2 creates ruptures / dense grains; T3 expands phase space. **TS:** three voices; SHIFT + pad adds rupture.

// KNOBS / AS

P	NORMAL	SHIFT
1	Grain emission rate	Grain overlap
2	Grain duration	Memory length / feedback
3	Melody base pitch	Phasing / orbit rate
4	Pulse shape / masking	Fold / saturation
5	Attractor / feedback	Melody > chaos / cuts
6	Tempo / release	Space / stereo / scale

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 pitch	Duration / overlap
2	T2 pitch	Formant colour
3	T3 pitch	Shape / fold
4	T1 decay	Memory / phaser
5	T2 decay	Chaos / mutation
6	T3 decay	Space / clock rate

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: mutate the current layer. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: priority step / trigger. **CV2:** smoothed continuous modulation; sends CC74.
Banana: 0-9 V; common GND.

// USB MIDI

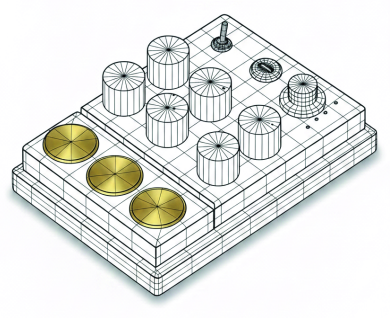
IN: channel 1 notes, transport, Clock; notes force an AS attack or open a TS voice. **OUT:** local / CV / pad notes and internal / CV Clock. **Panel:** CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

Start with moderate emission and overlap. CV owns clock priority; MIDI returns after 20 s without CV, internal after 30 s without an external source.

// generative instrument

Fracture

diceFriend / quick start



Mallets, bells and metal. Evolving phrases. Fractured rhythms.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: three evolving percussion parts play themselves. **TS:** play the three voice pads, CV or MIDI notes.

TS pads: three voices from the mallet / bell / metal instrument. Pitch follows evolving scales and phrases.

// SIX KNOBS / TWO LAYERS

P	NORMAL	SHIFT
1	Mallet: wood > glass > noise	Tempo: 7.5-375 BPM
2	Mallet decay	Density of all three parts
3	Bells: pure > FM > fold	Repeats: intensity / 2-6 hits
4	Bell feedback / sustain	Local rhythm deformation
5	Metal: bar > spring > noise	Glitch: freeze / reverse / bits
6	Global stereo feedback	Evolution / divisions

SHIFT: up = second layer; down = normal. **Pickup:** After switching layers or rolling DICE, turn the knob through its saved value to take control.

// DICE / SHIFT

Tap DICE: randomize the current layer only. **SHIFT up:** rhythm layer. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: clock; in TS, triggers all 3 voices.
CV2: timbre modulation; in TS, scale-step selection and rising-edge triggers.
Banana: 0-9 V; common GND.

// USB MIDI

IN: channel 1 notes, Clock and transport.
OUT: TS pad notes; 24 PPQN Clock with internal / CV timing. **Panel:** CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

High normal P6 sustains resonance; lower it to let sound fade. CV > MIDI > internal; MIDI returns after 20 s without CV, internal after 30 s without external timing.

// generative instrument

Arc_OS

diceFriend / quick start

Spectral phase distortion, four-operator FM, a pressure resonator and a centred sub interlock across six tuning worlds.

START: USB-C power + 3.5 mm stereo audio out. Begin with the listening level low.

AS: autonomous four-part rhythm. TS: T1 spectral / T2 FM / T3 pressure; T1 + T3 opens sub.

// SIX KNOBS / TWO LAYERS

P	NORMAL	SHIFT
1	Root: MIDI note 24-84	Tempo: 24-240 BPM
2	Arc: phase / spectral / FM	Stretch: spacing / FM ratios
3	Spread: melodic range	Memory: inertia / persistence
4	Contour: focus / envelope	Scale: 6 tuning worlds
5	Coupling: interaction	Glitch: capture / reverse / freeze
6	Decay: material sustain	Exciter: attack / brightness

// TUNING WORLDS

19 EDO subset / 24 EDO subset / 31 EDO subset / 13-step Bohlen-Pierce / septimal unequal / 22 EDO subset. ROOT transposes without pulling intervals toward 12-TET.
SHIFT: up = second layer; down = normal. Pickup: after changing layers, using DICE or receiving CC, move a knob to take control.

// DICE / MODES

Tap DICE: related mutation; root and tempo stay fixed. SHIFT + tap DICE: full mutation except root / tempo. Hold DICE 1.8 s: AS / TS. SHIFT + DICE 10 s: bootloader.

// PADS / CLOCK

AS pads add spectral, FM and pressure accents. TS pads hold those voices; T1 + T3 holds sub. Internal clock advances AS only. External CV or MIDI Clock can rotate through the four TS materials. Priority: CV > MIDI > internal; MIDI returns after 20 s, internal after 30 s.

// CV IN

CV1: priority trigger / clock; accepted periods generate USB Start + 24 PPQN Clock. CV2: smoothed material modulation and CC74. Banana: 0-9 V; common GND.

// USB MIDI

IN: ch 1-4 address spectral / FM / pressure / sub; ch 5-16 allocate free voices. Note On strikes; Note Off releases. OUT: four channels with Pitch Bend + Note On / Off. Panel CC20-31; CV2 CC74.

// FIRST PATCH

Choose 19 EDO or 24 EDO. Keep ARC / COUPLING below one third, GLITCH near zero and MEMORY / DECAY near the centre. Raise STRETCH, then add EXCITER and a little GLITCH.

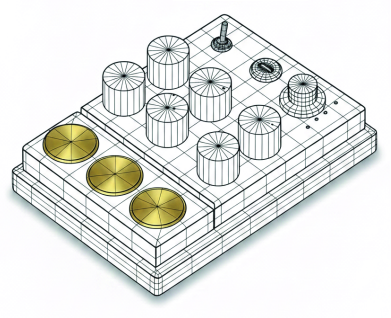
// LEDS

Layer LEDs show NORMAL / SHIFT. Lower LEDs 1-4 show spectral / FM / pressure / sub activity; DICE and mode changes flash mutation status.

// generative instrument

Gambi_OS

diceFriend / quick start



Slow veils, wet clatter and vinyl dust. Three drifting tracks melt into one soft, unstable wall.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: autonomous soundscape; each step moves one track. **TS:** pads open the tracks with slow swells.

AS: T1-T3 strike their track; hold for more texture. **TS:** T1-T3 open T1 / T2 / T3; release to fade.

// KNOBS / AS + TS

P	NORMAL	SHIFT
1	Level T1	Pitch T1: MIDI 36-60
2	Level T2	Pitch T2: MIDI 48-72
3	Level T3	Pitch T3: MIDI 60-84
4	Stickiness T1	Time: step 6 s to 1.2 s, glides
5	Stickiness T2	Instability: dissonance / wow / glitch
6	Stickiness T3	FX: wash; last quarter = excess

SHIFT: up = second layer; down = normal. **Pickup:** after changing layers, DICE or MIDI CC, move a knob about 8% to take control; it glides in over 200 ms.

// TRACKS / FX

Each track = **WELON** veil + **REZONANS** clatter and droplets + **FAKTURA** breath and grunts. Melodies walk a mode and hook on m2, tritone, M7 and m9. Chain: glitch, vinyl, tape wow, phaser, long reverb. **FX is level-matched:** more noise and crackle, not more volume.

// LEDS

Layer LEDs show NORMAL / SHIFT. LED 0 flashes on every step and blinks fast on AS / TS. LEDs 1-3 show T1 / T2 / T3 activity.

// DICE / SHIFT

Tap DICE: new values for the active layer and a new melody mode. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: priority clock; each edge moves the melody.
CV2: wetness: deeper wow, more clatter, grunts and glitches; sends CC74.
Banana: 0-9 V; common GND.

// USB MIDI

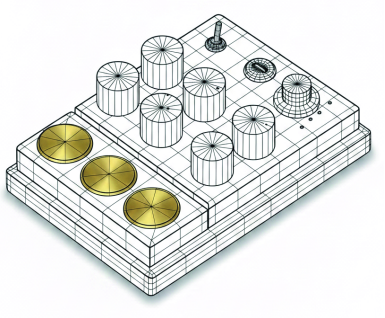
IN: ch 1-3 set pitch of T1-T3 and strike (TS: ch 1 plays all); Note Off releases. Clock 24 PPQN, Start / Stop. CC20-31 panel.
OUT: ch 16 notes, CC74, CC102-104 activity.

For wide wings, set Time low, FX around 70% and Instability near the centre. Push FX into its last quarter for vinyl excess.

// generative instrument

Absu_OS

diceFriend / quick start



Autonomous drums, bass, chords and a micro-sampler that tears its own output into fragments.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: evolving groove and self-sampling. **TS:** three performance layers with escalating holds.

T1: drums. **T2:** bass / chords. **T3:** memory / glitch. In TS, holds build repeated events, register jumps and cross-layer leakage.

// KNOBS / AS

P	NORMAL	SHIFT
1	Tempo: 42-200 BPM	Mutation rate
2	Density	Glitch / ratchets
3	Instability	Micro length / rate
4	Cycle / gear	Pitch / scale range
5	Percussion > tonal	Macro events
6	Space / mix	Degrade / reverse

// KNOBS / TS

P	NORMAL	SHIFT
1	Bass / chord pitch	Mutation rate
2	Held-event density	Glitch / ratchets
3	Escalation	Micro length / rate
4	Register range	Pitch / scale range
5	Percussion > tonal	Macro events
6	Space / mix	Degrade / reverse

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: new rhythmic organism. **SHIFT + tap:** more extreme variation. **DICE 1.8 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: trigger / clock and MIDI note / Clock output.
CV2: continuous modulation, CC74 / pressure; not a second clock.
Banana: 0-9 V; common GND.

// USB MIDI

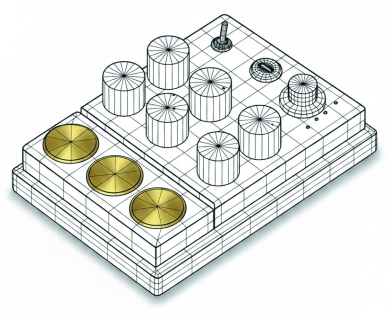
IN: channel 1 notes, transport and 24 PPQN Clock.
OUT: local notes, internal / CV Clock and CV2 messages. Incoming USB is not echoed. **Panel:** CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

Start with Density, Instability and Balance near halfway. Add SHIFT P2 Glitch and a short micro-fragment. Macro Void creates a deliberate brief silence.

// generative instrument

Gruz_OS

diceFriend / quick start



Bass, resonant noise and damaged ambience coupled through memory, feedback and saturation.

START: USB power + stereo audio out; begin with the listening level low. Starts in AS with all three tracks open.

AS: pads strike BASS / NOISE / AMBIENT and add density. **TS:** hold T1 / T2 / T3 to open the matching track.

// KNOBS / AS + TS

P	NORMAL	SHIFT
1	BASS level	BASS pitch: MIDI 24-60
2	NOISE level	NOISE pitch: MIDI 36-72
3	AMBIENT level	AMBIENT pitch: MIDI 36-72
4	BASS density / PM	MEMORY: recirculation / coupling
5	NOISE density	SATURATION
6	AMBIENT density / glitches	FX / MOTION / overload

Pickup: after startup, layer change, DICE or MIDI control, move a knob about 8% to take control; it glides for 200 ms.

// TRACKS / MODES

BASS: sub-rich PM. **NOISE:** pitched resonators with rungler excitation. **AMBIENT:** six detuned oscillators plus short clicks. Track energy cross-modulates the complete organism.

AS: all tracks remain open; clock gives a gentle bloom. **TS:** pads gate tracks; effect tails decay naturally. External clock can pulse all tracks until MIDI notes take over TS articulation.

// DICE / SHIFT

Tap DICE: randomize the active layer. **NS + DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader. Changing SHIFT cancels a hold.

// CV / CLOCK

CV1: priority trigger and step. **CV2:** smoothed coupling control; sends CC74, not V/oct.
Priority: CV, then MIDI, then internal. MIDI after 20 s without CV; internal after 30 s without external clock.

// USB MIDI

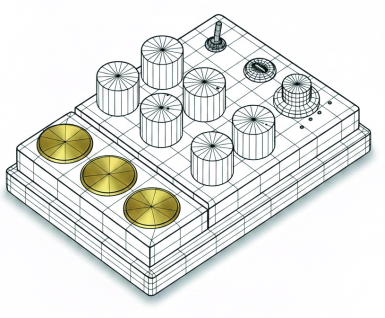
IN: AS ch 1-3 = tracks; TS ch 1 = all. Note On sets pitch and strikes; Note Off releases. Clock 24 PPQN; CC20-31 panel.
OUT ch 16: local notes, CV2 CC74 and activity CC102-104. No MIDI Thru or same-port echo.

LEDs: active layer, clock / step, then BASS / NOISE / AMBIENT activity. Start clean with low density; raise MEMORY, SATURATION and FX / MOTION gradually.

// generative instrument

SixOp_OS

diceFriend / quick start



The open-source diceFriend showcase: six-operator FM, three voices and eight algorithms.

START: USB power + stereo audio out; keep listening level low. Starts in AS: a three-voice drone. TS: three played voices.

// KNOBS / AS

P	NORMAL	SHIFT
1	Pitch: C1-C6	LFO depth
2	Chord: eight voicings	LFO rate
3	Algorithm: eight routings	Detune
4	Ratio: harmonic to metallic	Pulse: left = pure drone
5	Depth: FM index	Grit: lo-fi colour
6	Feedback	Space: stereo echo

// KNOBS / TS

P	NORMAL	SHIFT
1	Voice 1 pitch	Algorithm
2	Voice 2 pitch	Ratio
3	Voice 3 pitch	Depth
4	Voice 1 decay	Feedback
5	Voice 2 decay	Contour
6	Voice 3 decay	Detune

Pickup: move 8% from the armed position; takeover glides for 200 ms. AS Grit, Space and LFO Rate remain active in TS.

// PADS / DICE

AS pads: T1 Bright / T2 Metal / T3 Growl; touches re-strike. **TS:** T1-T3 play voices; hold for gate. Tine/Bell decay; Bass/Brass sustain.

NS tap: AS randomize NS; TS next Tine/Bell/Bass/Brass. **SH tap <1 s:** randomize SH. **NS hold 3 s:** AS/TS. **SH hold 10 s:** bootloader.

// CV / CLOCK

CV1: priority trigger. CV2: pulse clock or steady FM depth. CV > MIDI > internal: MIDI after 20 s without CV, internal after 30 s without external activity.

// USB MIDI

IN ch 1-15: notes, release, Clock 24 PPQN, transport; CC20-31 panel. OUT ch 16: local notes, panel CC, CV2 CC74; CV-derived Clock.

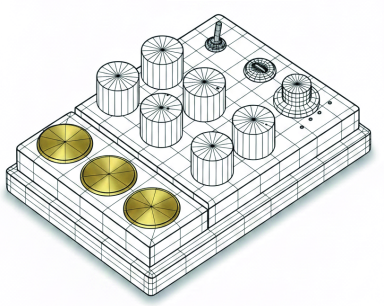
OPEN SOURCE: the complete commented reference implementation, renderer, developer notes and MIT license are available in the SixOp_OS GitHub package.

// generative instrument

Drzazga_OS

diceFriend / quick start

Kick and electric pulse hold the ground. Wood, triangle, can, blip and zap splinter in between, glitches fill the gaps.



START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: plays by itself; sparse by default, drill at the top of the knobs. **TS:** 32-step loop recorded with the pads, like PERC_OS.

AS: T1 kick, T2 hold = hat roll, T3 hold = freeze + avalanche. **TS:** T1 kick, T2 hat, T3 extra chosen by Material; hold 3 s = clear lane.

// KNOBS / AS + TS

P	NORMAL	SHIFT
1	Kick: deep and long to tight	Tempo: 50-200 BPM, 16th steps
2	Hat: dark tick to buzzing pulse	Density: sparse relabi to drill
3	Material: wood > tri > can > blip > zap	Crumbs: how often extras appear
4	Decay: extras dry to ringing	Relabi: time bending, phase shear
5	Glitch: grains in the gaps	Ratchet: repeats, rolls, drill
6	Space: room > drive > crush	Evolution: pattern rewrites itself

SHIFT: up = rhythm layer; down = timbre layer. **Pickup:** after changing layers, DICE or MIDI CC, move a knob about 8% to take control; it glides in over 200 ms.

// VOICES / GLITCH

Base: KICK + HAT stay stable; Density thins or fills the same groove. **Extras:** WOOD, TRI, CAN, BLIP (rungler), ZAP (arc) appear by chance on 5-13 step cycles, preferring the gaps. In TS they answer the recorded beat. **Glitch:** stutter, reverse, octave, crush, tape stop cut from the latest hits.

// LEDS

Layer LEDs show NORMAL / SHIFT. LED 0 flashes on every step, brighter on quarters; blinks fast on AS / TS. LEDs 1-3: kick, hat, extras + glitch.

// DICE / SHIFT

Tap DICE: new sounds for the extras; kick and hat stay. **SHIFT + tap:** new sequence (TS: variation of the loop). **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: priority clock; each edge = one 16th step.
CV2: level adds Density and Glitch; edges add an extra; sends CC74.
Banana: 0-9 V; common GND.

// USB MIDI

IN: ch 1 plays the T3 voice and sets its pitch; ch 2-8 voices 0-6; ch 10 GM drums. Clock 24 PPQN (6 = one 16th), Start / Stop. CC20-31 panel, CC117 remote DICE.
OUT: ch 16 GM notes, Clock, CC74.

For drill, push Density, Ratchet and Glitch above 80%. For floating splinters, keep Density low, Relabi high and Decay long.